

Module 7 – Advanced PHP Exercises

OOPs Concepts

Q.1 Define Object-Oriented Programming (OOP) and its four main principles: Encapsulation, Inheritance, Polymorphism, and Abstraction.

ans.1 Object-Oriented Programming (OOP) is a programming paradigm based on the concept of objects, which represent real-world entities. These objects contain data (attributes) and methods (functions) that operate on the data. OOP helps in organizing complex programs, promoting code reusability, modularity, and maintainability.

The four main principles of OOP are:

Encapsulation: It is the process of bundling data (variables) and methods (functions) that operate on that data into a single unit called a class.

It hides the internal details of an object and only exposes what is necessary through public methods.

Example: Using private variables and public getter/setter methods to control access.

Inheritance:

It allows a new class (child class or subclass) to acquire the properties and behaviors of an existing class (parent class or superclass).

Promotes code reusability and establishes a relationship between classes.

Example: A class Car can inherit from a class Vehicle.

Polymorphism:

It means “many forms” and allows one interface to be used for different data types or classes.

The same method name can behave differently based on the object that invokes it.

Example: A method draw() can perform differently for Circle, Rectangle, or Triangle objects.

Abstraction:

It focuses on hiding the complex implementation details and showing only the essential features of an object.

It simplifies code by allowing the programmer to interact with high-level operations instead of low-level details.

Example: A user interacts with a print() method without knowing how the printing process works internally.

Class

Q.2 Explain the structure of a class in PHP, including properties and methods.

ans.2 In PHP, a class is a blueprint for creating objects. It defines the properties (variables) and methods (functions) that describe the behavior and characteristics of those objects.

Structure of a Class in PHP

A typical PHP class structure includes:

Class Declaration

Properties (Attributes)

Methods (Functions)

Object Creation (Instantiation)

Syntax Example:

```
<?php
// Define a class
class Car {
```

```

// Properties (variables)
public $brand;
public $color;

// Constructor (special method to initialize properties)
public function __construct($brand, $color) {
    $this->brand = $brand;
    $this->color = $color;
}

// Method (function)
public function displayInfo() {
    echo "This car is a " . $this->color . " " . $this->brand . ".";
}
}

// Create an object (instance of the class)
$car1 = new Car("Toyota", "Red");

// Access method
$car1->displayInfo();
?>

```

Explanation of Components:

class Keyword:

Used to define a new class.

```
class Car { ... }
```

Properties:

Variables that store the data or state of an object.

Defined inside the class but outside any method.

Can have access modifiers like public, private, or protected.

```
public $brand;  
private $color;
```

Methods:

Functions that define the behavior or actions of an object.

Operate on the class's properties using \$this.

```
public function displayInfo() {  
    echo $this->brand;  
}
```

Constructor (__construct()):

A special method automatically called when an object is created.

Used to initialize object properties.

```
public function __construct($brand, $color) { ... }
```

Object Creation:

An object (instance) is created using the new keyword.

```
$car1 = new Car("Toyota", "Red");
```

Component	Description
Class	Blueprint for objects
Property	Variable inside a class

Method	Function inside a class
Constructor	Initializes object properties
Object	Instance of a class

Object

Q.3 What is an object in OOP? Discuss how objects are instantiated from classes in PHP

ans.3 What is an Object in OOP?

In Object-Oriented Programming (OOP), an object is an instance of a class.

A class acts as a blueprint or template, and an object is a real entity created from that blueprint.

Each object has:

Properties (Attributes): Represent the object's data or characteristics.

Methods (Functions): Define the actions or behaviors the object can perform.

For example:

If Car is a class, then a specific car like a red Toyota is an object of that class.

☞ Class = Template

☞ Object = Actual instance of that template

How Objects Are Instantiated from Classes in PHP

In PHP, objects are created (instantiated) from classes using the new keyword.

This process is known as instantiation.

Syntax:

```
$ObjectName = new ClassName();
```

If the class has a constructor, you can pass values to it:

```
$ObjectName = new ClassName(parameter1, parameter2);
```

Example:

```
<?php
```

```
// Define a class
```

```
class Car {
```

```
    public $brand;
```

```
    public $color;
```

```
    // Constructor method (called automatically when object is created)
```

```
    public function __construct($brand, $color) {
```

```
        $this->brand = $brand;
```

```
        $this->color = $color;
```

```
    }
```

```
    // Method to display car details
```

```
    public function displayInfo() {
```

```
        echo "This car is a " . $this->color . " " . $this->brand . ".";
```

```
    }
```

```
}
```

```
// Create (instantiate) objects from the class
```

```
$car1 = new Car("Toyota", "Red");
```

```
$car2 = new Car("Honda", "Blue");
```

```
// Access object methods
```

```
$car1->displayInfo(); // Output: This car is a Red Toyota.
```

```
echo "<br>";
```

```
$car2->displayInfo(); // Output: This car is a Blue Honda.
```

```
?>
```

Explanation:

class Car { ... } — Defines the class (blueprint).

new Car("Toyota", "Red"); — Creates an object (instance) of the Car class.

\$car1 and \$car2 — Are two different objects with their own property values.

\$this keyword — Refers to the current object within the class.

Key Points:

Concept	Description
Class	A blueprint or template for creating objects
Object	An instance of a class containing real data
Instantiation	The process of creating an object from a class
new keyword	Used to create objects in PHP
\$this keyword	Refers to the current object inside the class

Extends

Q.4 Explain the concept of inheritance in OOP and how it is implemented in PHP.

ans.4 Concept of Inheritance in Object-Oriented Programming (OOP)

Inheritance is one of the core principles of Object-Oriented Programming (OOP).

It allows a class (called a child or subclass) to inherit properties and methods from another class (called a parent or base class).

This promotes code reusability, extensibility, and a hierarchical class structure.

Key Points

The child class gets access to all public and protected members of the parent class.

The child can:

Use inherited properties/methods directly.

Override methods from the parent class.

Add its own new properties and methods.

The private members of the parent class are not accessible directly by the child class.

Implementation in PHP

PHP supports single inheritance, meaning a class can only inherit from one parent class (though it can use traits to include behavior from multiple sources).

Example:

```
<?php
```

```
// Parent class
```

```
class Animal {
```

```
    public $name;
```



```

        public function __construct($name) {
            $this->name = $name;
        }

        public function speak() {
            echo "$this->name makes a sound.";
        }
    }

    // Child class inheriting from Animal
    class Dog extends Animal {
        // Overriding parent method
        public function speak() {
            echo "$this->name barks!";
        }

        // New method specific to Dog
        public function fetch() {
            echo "$this->name is fetching the ball.";
        }
    }

    // Creating an object of the child class
    $dog = new Dog("Buddy");

    $dog->speak();    // Output: Buddy barks!

```

```
$dog->fetch();      // Output: Buddy is fetching the ball.
```

?>

□ Explanation

class Dog extends Animal — the Dog class inherits from the Animal class.

The constructor and speak() method are defined in Animal.

The Dog class:

Overrides the speak() method.

Adds a new method fetch().

When we create a Dog object, it can use both:

Inherited properties/methods (\$name, constructor)

Its own methods (fetch())

Modifier	Accessible in Parent	Accessible in Child	Accessible Outside
public	✓	✓	✓
protected	✓	✓	✗
private	✓	✗	✗

Using parent:: Keyword

The parent:: keyword allows a child class to access methods or constructors from its parent:

```
class Cat extends Animal {
```

```
public function __construct($name) {  
    parent::__construct($name); // Call parent constructor  
}  
  
    public function speak() {  
        parent::speak(); // Call parent method  
        echo " And $this->name meows.";  
    }  
}
```